How-To Library

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Data Replication Failover to an Oracle Data Guard Physical Standby Database

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Abstract

This article describes how to configure Data Replication to extract change data from an Oracle Data Guard primary database. This article also describes how to configure Data Replication to continue replication after a planned or unplanned outage causes the primary database to fail over to a physical standby database.

Supported Versions

- Data Replication 9.8.0

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Overview

Data Replication can extract change data from an Oracle primary database that is synchronized with a Data Guard physical standby database.

A physical standby database is an identical copy of the primary database. All of the database structures are identical to the primary database. Oracle uses Redo Apply to synchronize a physical standby database with the primary database. Oracle recovers the redo data that was received from the primary database and then applies the redo to the physical standby database.

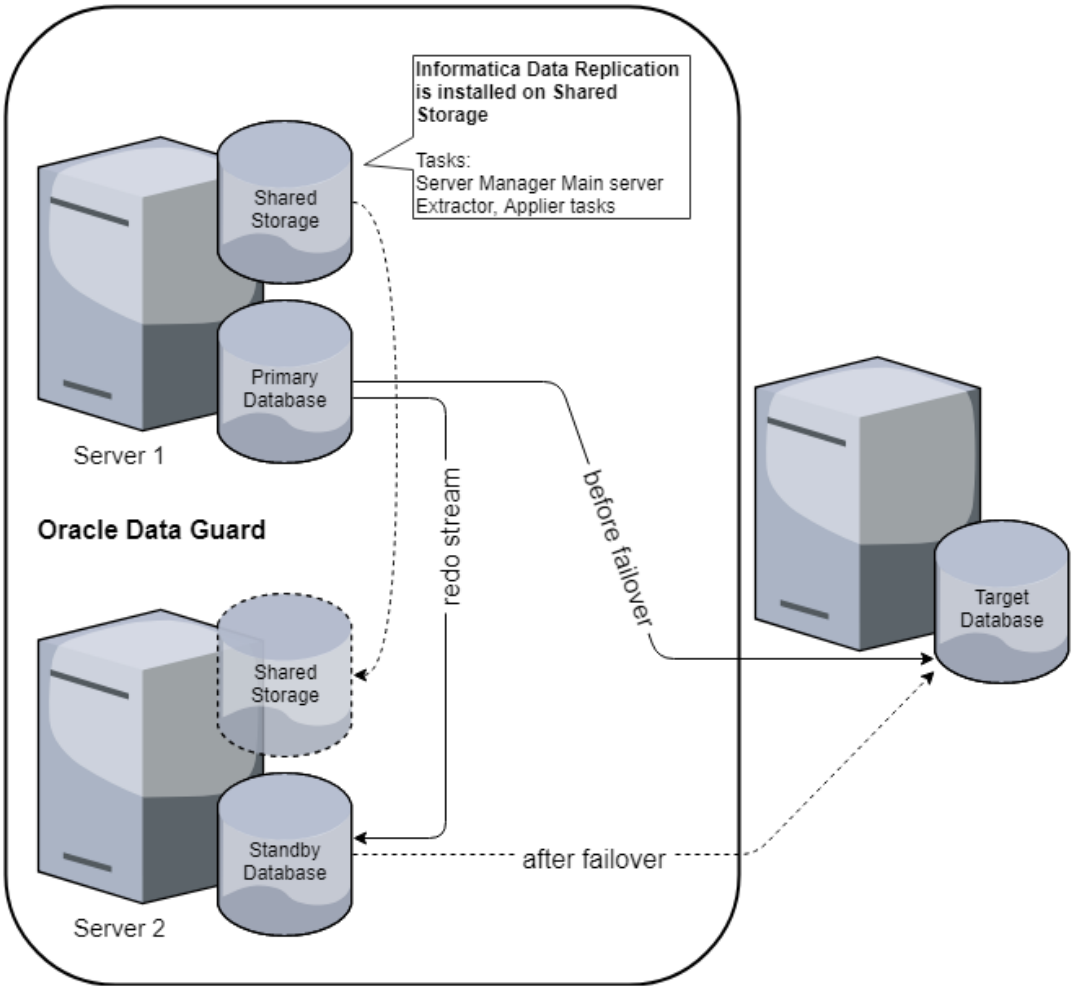
If a database outage occurs on the primary database from which Data Replication is extracting data and Oracle Data Guard fails over to a physical standby database, the standby database becomes the new primary database. Data Replication can resume extraction processing from the new primary database, provided that you complete the necessary configuration steps.

Topology

This article is based on a topology that includes one Oracle source primary database, one Data Guard standby database, and one target. The primary database is synchronized with the physical standby database. Data Replication extracts change data from the primary database.

Data Replication is installed in shared storage that is mounted to the Database Server 1. After Oracle Data Guard fails over to the physical standby database, the standby database on Server 2 becomes the primary database. You must configure Data Replication to continue change data replication from the new primary database on Server 2.

The following image shows this topology:



The Extractor, Server Manager, and Applier are installed in shared storage that is mounted to the primary database server on Server 1. The Data Replication Console can run on a standalone machine or in shared storage along with the Data Replication components.

The following table describes the characteristics of the Server 1 production system and the Server 2 physical standby systems:

Characteristic	Oracle Production Database	Standby Database
Server	10.51.40.26	10.51.40.27
Port	1521	1521
Database name	primary_database	primary_database
Instance name	primary_database	standby_database

The following considerations apply to this scenario:

- The Data Replication instance resides on a shared disk that is mounted to the primary database on Server 1. The required bandwidth depends on data volume. For best performance, install Data Replication on a shared storage device with the 10Gb iSCSI or 8/16Gb fibre channel (FC) interface.
- The Server Manager Main server has a database connection to primary database on Server 1. After a failover to the standby database, you must edit the database connection to point to the new primary database on Server 2.

Data Replication Licenses

To run the Server Manager on multiple Oracle Data Guard databases, you must request a license for each database from Informatica Global Customer Support. Associate each license to a database host name.

Tip: Store all of the licenses in a safe location that is different from the Data Replication installation directory. You cannot store both licenses in the same DataReplication.key file and use the same key file for both Oracle database instances.

Configuring Data Replication

In the Data Replication Console, create a configuration that replicates data from the Oracle primary database on Server 1 to the target.

1. Start the Data Replication Console.
2. To connect to the Server Manager Main server on the Oracle source server with the primary database, perform the following substeps:
   a. Click the **Server Manager** tab > **Configs** view.
   b. In the **Host** field, enter the IP address or host name of the primary database server. This example uses an IP address of 10.51.40.26.
   c. In the **Port** field, leave the default value of 8088.
   d. If the Server Manager Main server uses HTTPS connections, select **Use TLS**.
   e. In the **Login** field, leave the default value of idradmin.
   f. Leave the **Password** field empty.
3. Define connections to the primary database and the target. To define a connection, perform the following substeps:
   a. On the **Server Manager** tab > **Connections** view, click **New**.
   b. Enter database connection details. The following table describes these details with example values for primary database:

Field	Description
Name	Enter a database connection name.
Owner	Select the Data Replication user who is designated as the owner of the connection.
Type	Select the database type. For the primary database, the type is **Oracle**.

Field	Description
Host	Enter the IP address of a database. For the primary database, the example host IP address is 10.51.40.26.
Port	Specify the database port. For the primary database, use the default Oracle port number of 1521.
User	Enter the name of the database user.
Password	Enter the password of the database user.
Instance/DB	Enter the database instance name. For the primary database, use the name of primary_database.

c. Click **OK**.

4. To create a new replication configuration, click **New** and perform the following substeps:
   a. In the **New** dialog box, enter a configuration file name.
   b. In the **Owner** list, select the configuration owner.
   c. In the **Source Server Manager** list, select **Main**.
   d. In the **Target Server Manager** list, select the name of the subserver.
   e. Click **OK**.
5. On the **Source Database** tab > **Database Connection** view, select the primary database connection in the **Database Connection** list. Then click **Connect** to connect to the database.
6. On the **Target Database** tab, select the target connection in the **Database Connection** list. Then click **Connect**.
7. Generate target tables in the target database if you did not already create them. Perform the following substeps:
   a. Click **Schema** > **Generate Schema for the Target**.
   b. Under **Source/Target**, verify that the **Source** option is selected.
   c. In the **Schema/owner** field, enter the schema or owner name for the source tables that you want to use to generate the target tables.
   d. Select the source tables that you want to use to create corresponding target tables in the target database by selecting the **Convert** box next to each table.
   e. Under **Target**, in the **Database type** list, select the target database type.
   f. In the **Schema** field, select a schema name for the target tables.
   g. Click **Run**.
8. On the **Map Tables** tab, map the source and target tables by performing the following substeps:
   a. In the **Source Schema** list, select an Oracle source schema.
   b. Select a source table.
   c. In the **Target Schema** list, select the target schema.
   d. Select a target table.
   e. Click **SQL Apply**, **Audit Apply**, or **Merge Apply**, depending on the target type.
9. On the **Extract Range** tab, ensure that all directories with archived redo logs are specified correctly.
10. Save the configuration.

Running InitialSync

To materialize the target tables, run InitialSync.

To start the InitialSync task with the Data Replication configuration that you defined, perform the following steps:

1. On the **Server Manager** tab > **Configs** view, right-click the replication configuration row and click **Start Task** > **initialsync**.
2. When the InitialSync task completes, click **Close**.

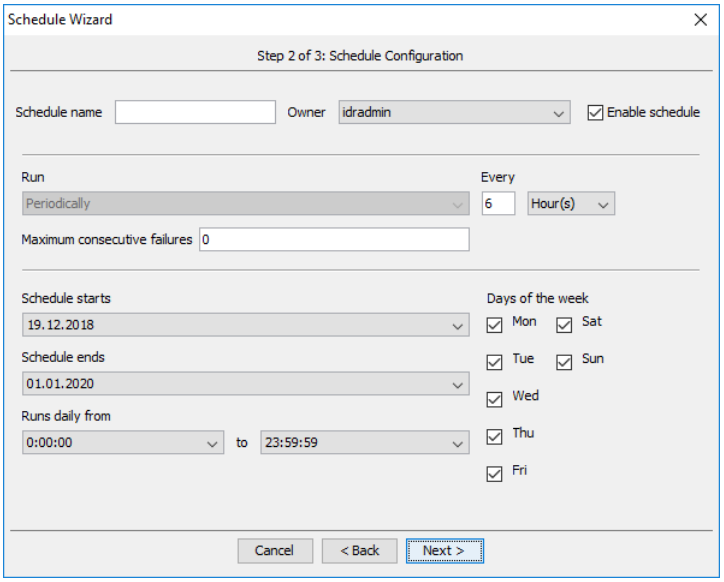
Creating a Replication Schedule

Create a schedule for running the change data replication tasks.

For convenience, use the **Schedule** wizard.

1. On the **Server Manager** tab > **Schedules** view, click **Wizard** on the toolbar to start the **Schedule** wizard.
2. On the **Schedule Type** page, select one of the following options:
   - **Sequential**. Defines one schedule that runs the Extractor, transmits the intermediate files, and runs the Applier sequentially.
   - **Parallel**. Defines two schedules that run independently. The first schedule runs the Extractor and transmits the intermediate files. The second schedule transmits the intermediate files and runs the Applier.
   - **Continuous**. Defines one schedule that runs the Extractor, transmits the intermediate files, and runs the Applier continuously for continuous replication.
3. Click **Next**.

   The **Schedule Configuration** page appears.



   Note: For the **Parallel** execution scheme, the **Extractor Schedule Configuration** page appears instead.
4. In the **Schedule name** field, enter a unique and descriptive name for the schedule.
5. In the **Owner** list, select a schedule owner.
6. Select **Enable schedule** to activate the schedule.

7. For the **Sequential** and **Parallel** execution schemes, perform the following substeps:
   a. Specify a repeat frequency under **Every**.
   b. In the **Maximum consecutive failures** field, specify the maximum number of consecutive times that a schedule can end with an error before the Server Manager disables the schedule.

      To allow an unlimited number of consecutive schedule failures, enter 0.
8. In the **Schedule starts** and **Schedule ends** fields, define the period during which the schedule runs. Click in each field to display a control for setting the date.

   For continuous replication, you can define a very long period, up to December 31, 2037.
9. Under **Days of the week**, select one or more days of the week when the tasks run.
10. If you want to limit a daily run to a specific time frame, set the **Runs daily from** fields.
11. For the **Parallel** execution scheme, click **Next** and enter the schedule options for the Applier schedule.
12. Click **Next**.

    The **Replication Configuration** page appears.
13. Select the configuration that you created for replicating data from the active standby database to the target and click **Next**.

    The wizard generates the schedule.
14. Click **Finish**.

    The **Schedules** view displays the schedule in the list of schedules and scheduled tasks.
15. To start the schedule, right-click the schedule row and click **Start**.

    The **Processes** box lists target processes for the task.

Failing Over to a New Primary Source Database

After a planned or unplanned outage causes the primary database from which Data Replication is extracting change data to stop and fail over to the physical standby database, you can configure Data Replication to resume change data extraction on the new primary database.

1. If the primary database on Server 1 is still available, perform the following substeps:
   a. Stop the replication schedule on the primary database server.
   b. Stop the Server Manager on the primary database server.

   Skip this step if the primary database on Server 1 failed.
2. Mount the Data Replication shared storage that contains the Data Replication components to Server 2.
3. Copy the license that is associated with the host name of standby database to the DBSYNC_HOME directory in shared storage, which is now mounted to Server 2. Copy the license under the name of DataReplication.key.
4. Start the Server Manager on Server 2.
5. From the Data Replication Console, connect to the Server Manager on Server 2 as the idradmin user.
6. Edit the Server Manager information. Perform the following substeps:
   a. On the **Server Manager** tab > **Servers** view, select the Server Manager Main server.
   b. Click **Edit**.
   c. In the **Host** field, enter the IP address of Server 2. In this scenario, the IP address is 10.51.40.27.
   d. Click **Save**.
7. Edit the database connection that you created for the primary database to point to the standby database.

8. Open the replication configuration for editing.
9. On the **Extract Range** tab, change the path to archived redo log files. Skip this step if the primary and standby servers use the same path to the archived redo logs.
10. Start the replication schedule that you stopped in Step 1.

Data Replication starts extracting data from the new primary database on Server 2.

Author

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